AVISHKAR MAHESH PAWAR

+91-8623005430 • avishkarpawar004@gmail.com • LinkedIn • GitHub • Portfolio • Credly

Professional Summary

Final-year Computer Science student specializing in distributed systems and fault-tolerant architectures. Built multiple distributed systems from scratch including a key-value database with Raft consensus and an event streaming platform with partition replication. Deep understanding of consistency protocols, leader election, and data sharding. Strong background in C++, Java, and system design with hands-on experience in building scalable, highly available systems.

Education

MIT Academy of Engineering, Pune 2022 - 2026

B.Tech. - Computer Engineering — CGPA: 9.24/10

Projects

MiniDB — Distributed Database with ACID Transactions

C++17, Boost.Asio, Protocol Buffers, CMake

- Built Raft consensus algorithm for leader election and log replication, achieving fault tolerance across 3-node cluster with automatic failover in <2 seconds
- Implemented consistent hashing for data sharding across multiple nodes with HTTP monitoring dashboard for real-time cluster health tracking
- Designed write-ahead logging (WAL) for crash recovery with REDO/UNDO operations, ensuring durability with sub-second recovery time

Event Streaming Platform

Java 17, Spring Boot, Netty, Apache Kafka Protocol

- Designed partition replication protocol with leader-follower architecture for fault tolerance and high availability
- Implemented offset management with configurable retention policies and compaction strategies
- Built cluster coordination using ZooKeeper for leader election and metadata management

AI-Powered Vector Database

Python 3.11, FastAPI, NumPy, Numba

- Implemented graph-based indexing (HNSW) with hierarchical layers for sub-linear search complexity O(log n) on 1,000,000+ vectors
- Designed sharding mechanism with consistent hashing and write-ahead logging for horizontal scalability and crash recovery
- Built RESTful API using FastAPI with async endpoints handling 1000+ concurrent requests

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript
Frameworks: Spring Boot, FastAPI, React, Node.js, Express.js
AI/ML: TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba
Databases: PostgreSQL, MySQL, MongoDB, Redis, Custom Database Implementations
Tools: Git, Docker, Linux, AWS, CMake, Maven, Prometheus
Distributed Systems: Kafka, gRPC, Protocol Buffers, Raft Consensus, Consistent Hashing
CS Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design

Experience

OSMOS Jul 2025 - Present

Software Developer

- Worked on backend services using Node.js and developed applications using Python
- Contributed to product development from scratch, including designing, implementing, and testing core features

Core-Decimal Solutions Feb 2025 - Aug 2025

Software Developer

- Contributed to full-stack project using Vue.js, Express.js, and Sequelize with MVC architecture
- Developed reusable Vue.js components and integrated RESTful APIs with Axios

Achievements & Certifications

GATE 2026: Secured AIR 3124 with GATE Score 601 and 51.45/100 marks in CS — 3x rank improvement from GATE 2025 (AIR 9860)

GATE 2025: Qualified with AIR 9860, GATE Score: 453, Marks: 39.32/100

First Place in Technodium 25: Won first place in technical competition

Certifications: CCNA: Introduction to Networks, CCNA: Switching, Routing, and Wireless Essentials, CCNA: Enterprise Networking, Security, and Automation

Extra-Curricular Activities

Outdoor Sports & Games: Active participant in outdoor sports and recreational activities